

Q1.

Two smooth spheres A and B of equal radius are free to move on a smooth horizontal surface.

The masses of A and B are m and $4m$ respectively.

The coefficient of restitution between the spheres is e .

The spheres are projected directly towards each other, each with speed u , and subsequently collide.

- (a) Show that the speed of B immediately after the impact with A is

$$\frac{u(3-2e)}{5} \quad (4)$$

- (b) Find the speed of A in terms of u and e .

(2)

- (c) Comment on the direction of motion of the spheres after the collision, justifying your answer.

(2)

- (d) The magnitude of the impulse on B due to the collision is I .

Deduce that

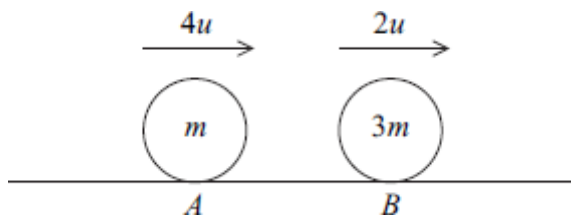
$$\frac{8mu}{5} \leq I \leq \frac{16mu}{5} \quad (3)$$

(Total 11 marks)

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Q2.

A smooth sphere A , of mass m , is moving with speed $4u$ in a straight line on a smooth horizontal table. A smooth sphere B , of mass $3m$, has the same radius as A and is moving on the table with speed $2u$ in the same direction as A .



The sphere A collides directly with sphere B . The coefficient of restitution between A and B is e .

- (a) Find, in terms of u and e , the speeds of A and B immediately after the collision.

(6)

- (b) Show that the speed of B after the collision cannot be greater than $3u$.

(2)

- (c) Given that $e = \frac{2}{3}$, find, in terms of m and u , the magnitude of the impulse exerted on B in the collision.

(3)

(Total 11 marks)

Q3.

A ball is dropped from a height of 2.5 m above a horizontal floor. The ball bounces repeatedly on the floor.

- (a) Find the speed of the ball when it first hits the floor.

(2)

- (b) The coefficient of restitution between the ball and the floor is e .

- (i) Show that the time taken between the first contact of the ball with the floor and the second contact of the ball with the floor is $\frac{10e}{7}$ seconds.

(3)

- (ii) Find, in terms of e , the time taken between the second contact and the third contact of the ball with the floor.

(1)

- (c) Find, in terms of e , the total vertical distance travelled by the ball from when it is dropped until its third contact with the floor.

(5)

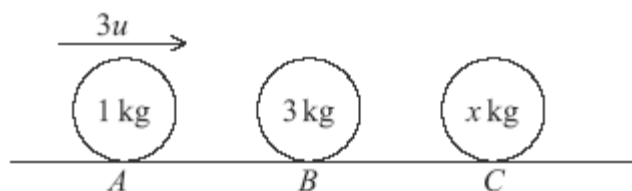
- (d) State a modelling assumption for answering this question, other than the ball being a particle.

(1)

(Total 12 marks)

Q4.

Three smooth spheres, A , B and C , of equal radii have masses 1 kg, 3 kg and x kg respectively. The spheres lie at rest in a straight line on a smooth horizontal surface with B between A and C . The sphere A is projected with speed $3u$ directly towards B and collides with it.



The coefficient of restitution between each pair of spheres is $\frac{1}{3}$.

- (a) Show that A is brought to rest by the impact and find the speed of B immediately after the collision in terms of u .

(6)

- (b) Subsequently, B collides with C .

Show that the speed of C immediately after the collision is $\frac{4u}{3+x}$.

Find the speed of B immediately after the collision in terms of u and x .

(6)

- (c) Show that B will collide with A again if $x > 9$.

(2)

- (d) Given that $x = 5$, find the magnitude of the impulse exerted on C by B in terms of u .

(2)

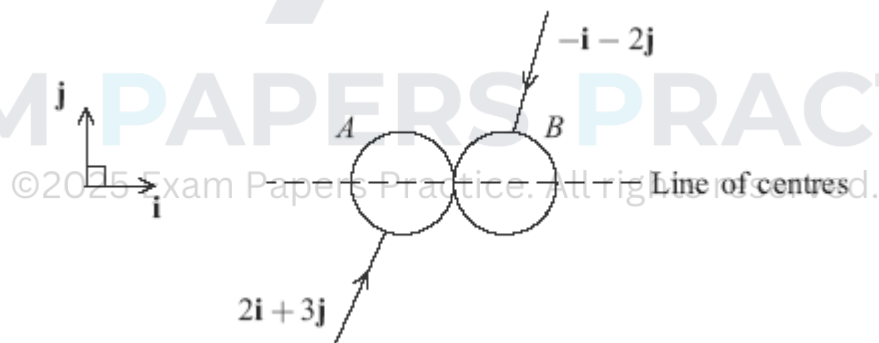
(Total 16 marks)

Q5.

Two smooth spheres, A and B , have equal radii and masses 1 kg and 2 kg respectively.

The sphere A is moving with velocity $(2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ and the sphere B is moving with velocity $(-\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$ on the same smooth horizontal surface.

The spheres collide when their line of centres is parallel to the unit vector $\mathbf{i}$, as shown in the diagram.



- (a) Briefly state why the components of the velocities of A and B parallel to the unit vector $\mathbf{j}$ are not changed by the collision.

(1)

- (b) The coefficient of restitution between the spheres is 0.5.

Find the velocities of A and B immediately after the collision.

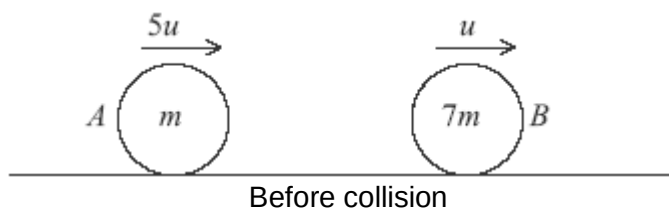
(6)

(Total 7 marks)

Q6.

A smooth sphere A of mass m is moving with speed $5u$ in a straight line on a smooth

horizontal table. The sphere A collides directly with a smooth sphere B of mass $7m$, having the same radius as A and moving with speed u in the same direction as A . The coefficient of restitution between A and B is e .



- (a) Show that the speed of B after the collision is $\frac{u}{2}(e+3)$. (5)

- (b) Given that the direction of motion of A is reversed by the collision, show that $e > \frac{3}{7}$. (4)

- (c) Subsequently, B hits a wall fixed at right angles to the direction of motion of A and B . The coefficient of restitution between B and the wall is $\frac{1}{2}$. Given that after B rebounds from the wall both spheres move in the same direction and collide again, show also that $e < \frac{9}{13}$. (4)

(Total 13 marks)

Q7.

Two small smooth spheres, A and B , of equal radii have masses 0.3 kg and 0.2 kg respectively. They are moving on a smooth horizontal surface directly towards each other with speeds 3 m s^{-1} and 2 m s^{-1} respectively when they collide. The coefficient of restitution between A and B is 0.8 .

- (a) Find the speeds of A and B immediately after the collision. (6)
- (b) Subsequently, B collides with a fixed smooth vertical wall which is at right angles to the path of the sphere. The coefficient of restitution between B and the wall is 0.7 .

Show that B will collide again with A .

(3)
(Total 9 marks)

Q8.

Three smooth spheres A , B and C of equal radii and masses m , m and $2m$ respectively lie at rest on a smooth horizontal table. The centres of the spheres lie in a straight line with B between A and C . The coefficient of restitution between any two spheres is e .

The sphere A is projected directly towards B with speed u and collides with B .

- (a) Find, in terms of u and e , the speed of B immediately after the impact between A and B .

(5)

- (b) The sphere B subsequently collides with C . The speed of C immediately after this collision is $\frac{3}{8}u$. Find the value of e .

(7)

(Total 12 marks)

Q9.

Three smooth spheres, A , B and C , of equal radii lie at rest in a straight line on a smooth horizontal table. Sphere A has mass $4m$ and is set in motion with velocity u . It collides directly with sphere B which has mass $2m$. The coefficient of restitution between A and B is e .

- (a) Find, in terms of e and u , the velocities of A and B immediately after the collision.

(5)

- (b) Given that the magnitude of the impulse exerted by A on B is $\frac{12mu}{5}$, show that

$$e = \frac{4}{5}$$

(3)

- (c) After the collision between A and B , the sphere B collides directly with the sphere C which has mass m . The coefficient of restitution between B and C is $\frac{4}{5}$.

Find, in terms of u , the velocity of B after this collision.

(4)

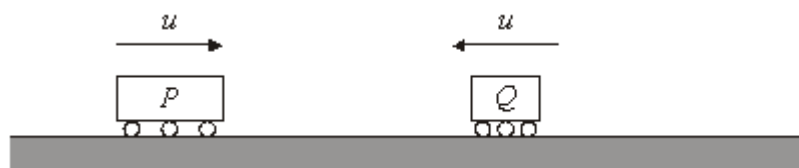
- (d) Determine whether or not A and B will collide again. Give a reason for your answer.

(2)

(Total 14 marks)

Q10.

Two toy train carriages, P and Q , have masses $2m$ and m respectively. They are moving directly towards each other, as shown in the diagram. Both carriages have speed u .



The carriages collide and subsequently Q moves in the same direction as P . The

coefficient of restitution between P and Q is e .

- (a) (i) Show that the speed of carriage P immediately after the collision is $\frac{2u}{3}(1 - 2e)$ and find the speed of carriage Q . (7)

- (ii) Deduce that $e < \frac{1}{2}$. (2)

- (b) The magnitude of the impulse on P due to the collision is I . Show that:

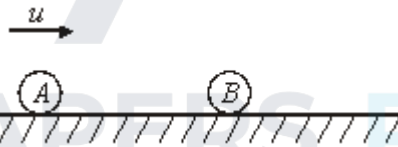
- (i) $I = \frac{4mu}{3}(1 + e)$; (3)

- (ii) $\frac{4mu}{3} \leq I < 2mu$. (2)

(Total 14 marks)

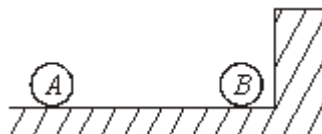
Q11.

Two identical smooth spheres, A and B , are each of mass m . Sphere A moves with speed u on a smooth horizontal floor directly towards sphere B , which is at rest. The spheres are shown in the diagram.



The coefficient of restitution between A and B is e .

- (a) Show that the speed of B immediately after the impact with A is $\frac{1}{2}u(1 + e)$ and find the speed of A , in terms of u and e . (6)
- (b) Sphere B constantly collides with a vertical wall which is perpendicular to the direction of motion of B .



The coefficient of restitution between B and the wall is $\frac{2}{3}$.

State, in terms of u and e , the speed of B immediately after it hits the wall.

(1)

- (c) After B has hit the wall, A and B are moving directly towards each other with the same speed.

- (i) Show that e , the coefficient of restitution between A and B , is $\frac{1}{5}$.

(2)

- (ii) State the speed of A and B in terms of u .

(1)

- (d) The first collision between A and B occurs at a distance d from the wall. By modelling the spheres as particles, determine, in terms of d , the distance of the second collision from the wall.

(6)

(Total 16 marks)

Q12.

Two cars, A and B , are travelling in the same direction on a straight horizontal road. Car A is travelling at 12 m s^{-1} when it collides with car B which is travelling at 8 m s^{-1} , as shown in the diagram.



Car A has mass 800 kg and car B has mass 1000 kg . To model this collision, the cars can be considered as particles. The coefficient of restitution between these particles is $\frac{1}{8}$.

Show that the speed of car B immediately after the collision is 10 m s^{-1} and find the speed of car A .

(Total 7 marks)

Q13.

In an accident, a car A collides directly with a stationary car B . The masses of A and B are 1200 kg and 1500 kg respectively. The speed of A immediately before it collides with B is 43.2 km h^{-1} . To model the collision, the cars may be considered as particles. The coefficient of

restitution between these particles is $\frac{1}{4}$.

- (a) Show that the speed of B immediately after the collision is 24 km h^{-1} and find the speed of A .

(7)

- (b) Show that the magnitude of the impulse of A on B due to the collision is $10\,000\text{ N s}$. (3)
- (c) During the collision a varying force, F newtons, acts between the cars. In a simple model, F can be assumed to increase uniformly from zero to a maximum value and then decrease uniformly to zero.
- (i) Sketch a graph showing how F varies with time, t seconds. (2)
- (ii) Given that the cars are in contact for 0.01 seconds, use your graph to find the maximum value of F . (3)
- (iii) State **one** criticism of this model of the force F . (1)
- (Total 16 marks)

Q14.

A particle P , of mass $6m$, travelling with speed $3u$ collides directly with a particle Q , of mass $2m$, travelling in the opposite direction with speed u .

The coefficient of restitution between the two particles is $\frac{1}{4}$.

- (a) Show that the speed of P after the collision is $\frac{7}{4}u$. (6)
- (b) Find the speed of Q after the collision. (2)
- (Total 8 marks)

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Q15.

Three particles, P , Q and R , of masses m , $3m$ and $5m$ respectively, are in a straight line on a smooth horizontal surface. Initially Q and R are at rest and P is travelling with velocity $2u$ directly towards Q . When the distance between P and Q is a , particle Q is struck a blow which results in Q travelling with the same velocity as P .



- (a) Find the impulse of the blow. (2)
- (b) In the subsequent motion, Q collides directly with R . In this collision, Q is brought to rest.
- (i) Find the velocity of R after the collision. (3)

(ii) Show that the coefficient of restitution between Q and R is $\frac{3}{5}$.

(2)

(iii) Find the time after this collision when P and Q collide. Give your answer in terms of a and u .

(2)

(c) Another two particles, A and B , of masses m and M respectively, are on the smooth horizontal surface. Particle A is travelling with velocity u directly towards B , which is at rest. The two particles collide. Particle A is brought to rest by this collision.

Prove that $M \geq m$.

(3)

(Total 12 marks)

Q16.

A particle, A , of mass $4m$ moving with constant speed u collides directly with a stationary particle, B , of mass m . After the collision both particles move in the same direction. The velocity of B is twice that of A .

(a) Show that the speed of the $4m$ mass after the collision is $\frac{2}{3}u$.

(4)

(b) Find the coefficient of restitution between the particles in the collision.

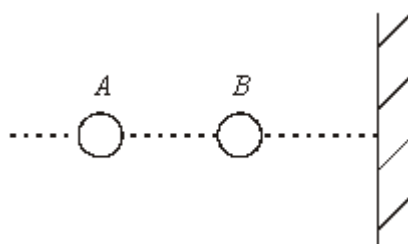
(5)

(Total 9 marks)

Q17.

Two spheres, A of mass $2m$ and B of mass m , are at rest on a smooth horizontal surface.

The centres of the spheres lie on a straight line which is perpendicular to a fixed smooth vertical wall. All motion takes place in the direction of this line.



Sphere A is hit by a blow of impulse J and moves with a speed of $3u$ directly towards the wall. Model the spheres as particles.

(a) Find J in terms of m and u .

(2)

(b) Sphere A collides directly with sphere B which is at rest.

The coefficient of restitution in this collision is $\frac{3}{5}$.

Find the speeds of A and B after this impact in terms of u .

(6)

- (c) Sphere B now hits the wall.

The coefficient of restitution between the sphere and the wall is $\frac{1}{2}$.

Find the speed of B after it strikes the wall.

(2)

- (d) Initially sphere B is a distance $4a$ from the wall. Find the distance of B from the wall when B and A collide again.

(7)

(Total 17 marks)



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